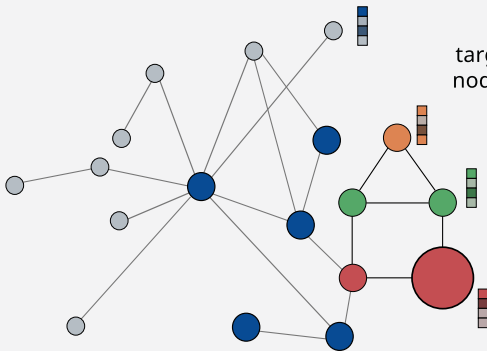


Clexplainer:

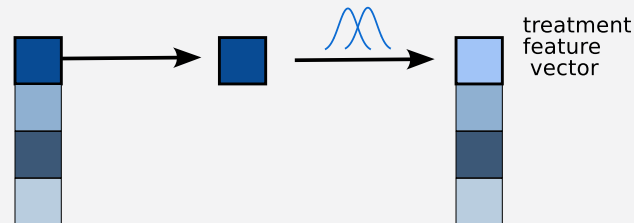
L-hop neighbourhood of target node

feature vector
vector
target explanation
node



Counterfactual generation:

for each feature in the node:

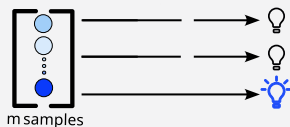


calculate causal effect:

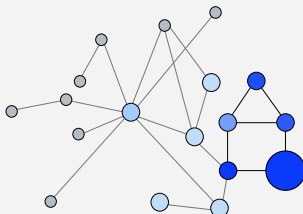
$$\tau_{w,i}(a) = Y_{t(a)}(\mathcal{G}, v, w, i) - Y_c(\mathcal{G}, v, w, i).$$

Explanation Subgraph:

take maximum Causal effect for each node:

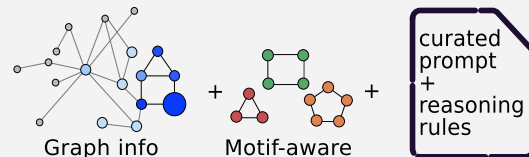


choose top-k causal effect nodes as part of explanation subgraph:



estimated causal effect

Textual Explanation:



=

Textual explanation of the explanation subgraph for a given node. Only positive nodes, or in the multi-class case (1,2,3) have an explanation sub-graph. Nodes in blue do not have an explanation sub-graph. A similar procedure occurs for graph classification and link prediction, such that in the graph classification case there is one explanation sub-graph per graph. And for link prediction, the explanation sub-graph exist for existing links.